

The Dangers of Change Approval Processes

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Abstract

Change approval processes have historically served as a central mechanism for controlling modifications to production systems, ensuring compliance, and maintaining operational stability. Traditional frameworks such as ITIL emphasize structured change enablement, formal approvals, and Change Advisory Boards (CABs). Modern DevOps research including findings from Puppet's DevOps Reports, Google's DORA studies, Atlassian's ITSM guidance, and NIST's Cybersecurity Framework reveal that these processes often introduce significant risks. These dangers include delivery delays, symbolic approvals, weakened segregation of duties, increased operational exposure, reduced auditability, and the normalization of unsafe practices. This paper analyzes the dangers of change approval processes using verified, working references and explains how these risks manifest in DevOps and software engineering environments. Recommendations for modernizing change governance are also provided.

1. Introduction

Change approval processes are widely used across industries to control modifications to production systems, infrastructure, and software. Historically, these processes were designed to prevent unauthorized changes, ensure compliance, and reduce risk. ITIL's Change Enablement practice (PeopleCert) emphasizes structured approvals, risk assessment, and CAB reviews to maintain service stability.

However, modern DevOps operate at high velocity, deploying changes multiple times per day. Research from Puppet and DORA shows that manual approvals often slow delivery, reduce deployment frequency, and fail to improve stability. Atlassian's ITSM guidance further notes that CABs frequently become bottlenecks rather than safeguards.

This paper examines the dangers of change approval processes and explains how these issues apply in DevOps and software engineering.

2. Purpose of Change Approval Processes

Change approval processes were created to serve foundational governance, operational, and security functions. Based on ITIL 4 Foundation (PeopleCert), Atlassian's ITSM guidance, and NIST's governance principles, the key functions are:

1. Risk Mitigation

ITIL 4 Foundation emphasizes that change enablement exists primarily to reduce risk. Before a change is deployed, organizations evaluate potential impacts on services, infrastructure, security, and users. This includes assessing whether the change could cause downtime, data loss, or instability.

2. Compliance Assurance

NIST's Cybersecurity Framework highlights the importance of governance and adherence to regulatory requirements. Change approvals ensure that modifications follow internal policies, external regulations, and industry standards.

3. Quality Control

Atlassian explains that approvals verify whether testing, validation, and peer review have been completed. This ensures that changes meet quality standards before reaching production.

4. Accountability and Traceability

ITIL stresses documenting who approved a change, when it was approved, and why. This creates a traceable record for audits or incident investigations.

5. Prevention of Unauthorized Changes

NIST emphasizes controlled access to production systems. Approvals ensure that only authorized individuals can modify critical environments, preventing accidental or malicious changes.

6. Coordination Across Teams

Atlassian notes that change management facilitates communication between development, operations, security, and business stakeholders. Approval ensures that all impacted groups are aware of upcoming changes.

7. Change Prioritization

ITIL explains that changes vary in urgency and importance. Approval processes help organizations prioritize changes based on business value, risk, and timing.

8. Segregation of Duties (SoD)

NIST and ITIL emphasize separating development and approval roles. This prevents developers from approving their own changes, reducing bias and ensuring independent review.

9. Documentation and Recordkeeping

ITIL requires organizations to maintain detailed records of changes, approvals, and related evidence. Documentation supports audits, compliance checks, and post-incident analysis.

10. Operational Stability and Predictability

Approvals provide structure and predictability. By reviewing changes before deployment, organizations reduce unexpected disruptions and maintain stable operations.

3. Structural Dangers of Change Approval Processes

Using verified information from Puppet, DORA, Atlassian, NIST, and ITIL, the eight major dangers are:

1. Symbolic Approvals (Rubber-Stamping)

Atlassian reports that CABs often approve changes without reviewing evidence thoroughly. Approvers may skim summaries or rely on trust rather than verification. This creates a false sense of security and undermines the purpose of change management.

2. Bottlenecks and Delivery Delays

Puppet's DevOps Reports show that organizations with manual approvals deploy less frequently and have longer lead times. DORA confirms that manual approvals slow down delivery without improving stability.

3. Breakdown at Scale

Atlassian explains that CABs cannot scale to environments with hundreds of changes per week. As organizations grow, manual approvals become unmanageable, leading to rushed decisions and increased operational risk.

4. Loss of Segregation of Duties

ITIL stresses separating development and approval roles. In small teams or fast-paced environments, developers often approve their own changes, eliminating independent review and increasing risk.

5. Increased Likelihood of Unauthorized or Unsafe Changes

NIST highlights the need for strong governance and oversight. Weak approval processes allow changes to bypass testing or security checks, leading to configuration drift and vulnerabilities.

6. Erosion of Auditability and Traceability

Symbolic approvals produce weak audit trails. NIST requires defensible evidence for governance. When approvals lack documentation or justification, organizations cannot demonstrate compliance.

7. Human Error and Cognitive Overload

Puppet's research shows that humans frequently make mistakes under pressure. Approvers must evaluate complex evidence across multiple tools, often with limited context. Cognitive overload leads to oversight and inconsistent decisions.

8. Normalization of Deviance

Atlassian notes that emergency changes and informal approvals often become routine. Over time, teams normalize risky behavior, increasing long-term operational exposure.

4. How These Dangers Apply in DevOps and Software Engineering

4.1 Conflict with DevOps Principles

DevOps emphasizes automation, continuous delivery, rapid feedback loops, and shared responsibility. Manual approvals conflict with these principles by introducing slow, centralized decision-making.

4.2 Incompatibility with CI/CD Pipelines

DORA research shows that high-performing DevOps teams rely on automated testing and deployment. Manual approvals interrupt pipelines, increasing batch size and risk.

4.3 Increased Risk in Microservices and Cloud Environments

Modern architecture involves distributed systems and frequent deployments. Atlassian notes that CABs cannot keep pace with cloud-native environments.

4.4 Reduced Developer Productivity

Puppet's reports show that waiting for approvals causes context switching, reduced focus, and lower morale.

4.5 Weak Security Assurance

NIST emphasizes automated controls and continuous monitoring. Manual approvals do not provide meaningful security validation.

5. Case Studies and Industry Evidence

5.1 Puppet: DevOps Report Findings

Puppet's research shows that organizations with manual approvals deploy less frequently and have longer lead times. High performers automate approvals entirely.

5.2 DORA: Deployment Frequency and Stability

DORA confirms that manual approvals do not improve stability. Automated pipelines with strong testing produce better outcomes.

5.3 Atlassian: CAB Inefficiencies

Atlassian reports that CABs often delay changes without reducing incidents, making them ineffective risk controls.

5.4 NIST: Governance Requirements

NIST highlights the need for defensible evidence and automated governance, which manual approvals rarely provide.

6. Recommendations

Organizations should:

1. Replace manual approvals with automated, evidence-based gates.
2. Implement risk-based workflows.
3. Use policy-as-code for compliance.
4. Maintain segregation of duties through automation.
5. Unify evidence across tools.
6. Monitor for normalization of deviance.
7. Reduce reliance on CABs in high-velocity environments.

7. Conclusion

Change approval processes were designed to reduce risk, but modern evidence shows they often create new dangers. Puppet, DORA, Atlassian, NIST, and ITIL all highlight the limitations of

manual approvals in modern environments. As systems scale and delivery accelerate, organizations must adopt automated, evidence-based, and risk-driven approaches to ensure safety, compliance, and operational resilience.

References

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